

j4_talk -- pin diagram

Teensy 4.1 + Audio Shield Rev D, component side up, USB at the TOP

		+===== [USB] =====+					
		GND	-	GND	5V	-	5V
		0	-	0	GND	-	GND
		1	-	1	3.3V	-	3.3V
front_LED_01	(T5)	2	*-	2	23	-	(audio I2S)
front_LED_02	(T4)	3	*-	3	22	-*	front_LED_09 (T4)
front_LED_03	(T3)	4	*-	4	21	-	(audio I2S)
front_LED_04	(T2)	5	*-	5	20	-	(audio I2S)
front_LED_05	(T1)	6	*-	6	19	-	(audio I2C)
(audio)		7	-	7	18	-	(audio I2C)
(audio)		8	-	8	17	-*	front_LED_08 (T3)
front_LED_06	(T1)	9	*-	9	16	-*	front_LED_07 (T2)
side_LED_07	(T5)	10	*-	10	15	-*	side_LED_01 (T2)
side_LED_08	(T5)	11	*-	11	14	-*	front_LED_10 (T5)
		12	-	12	13	-	(LED_BUILTIN)
Serial6 TX		24	*-	24	41	-	
Serial6 RX		25	*-	25	40	-	
		26	-	26	39	-	
		27	-	27	38	-	
side_LED_02	(T2)	28	*-	28	37	-*	side_LED_06 (T4)
side_LED_03	(T3)	29	*-	29	36	-*	side_LED_05 (T4)
		30	-	30	35	-	
		31	-	31	34	-	
		32	-	32	(bottom: SD card + SMD pads)		
side_LED_04	(T3)	33	*-	33			
		+=====+					

Serial6: 24 (TX) -> j4_receiver GPIO 2 ; 25 (RX) <- j4_receiver GPIO 17
 Audio Shield Rev D uses 7, 8, 20, 21, 23 (I2S) + 18, 19 (I2C) + built-in SD.
 front_LED_01..10 = front row left to right (pins 2,3,4,5,6,9,16,17,22,14);
 front 05/06 light first. side_LED_01..08 = down the side, top to bottom
 (pins 15,28,29,33,36,37,10,11).

Pin(s)	Function
2, 3, 4, 5, 6	front_LED_01..05 -- front row left half (tiers 5, 4, 3, 2, 1)
9, 16, 17, 22, 14	front_LED_06..10 -- front row right half (tiers 1, 2, 3, 4, 5)
6, 9	front_LED_05 + 06 = innermost pair, FIRST to light (tier 1)
15, 28	side_LED_01, side_LED_02 -- top of the side column (tier 2)
29, 33	side_LED_03, side_LED_04 (tier 3)
36, 37	side_LED_05, side_LED_06 (tier 4)
10, 11	side_LED_07, side_LED_08 -- bottom of the side column (tier 5)
24	Serial6 TX to j4_receiver (receiver GPIO 2)
25	Serial6 RX from j4_receiver (receiver GPIO 17)
7, 8, 20, 21, 23	Audio Shield Rev D I2S (do not reuse)
18, 19	Audio Shield Rev D I2C (do not reuse)